

Assignment 3

The Anatomy of a Regression Model and Generalized Linear Models (Weeks 5–6)

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Due: Monday, October 26, 2026

1 Overview

This assignment covers material from Week 5 (The Anatomy of a Regression Model) and Week 6 (Generalized Linear Models). It is worth **5 points**, allocated for completion, and is due **Monday, October 26**. You may resubmit as many times as you like over the course of the semester. You are encouraged to work in groups, but each of you must submit your own work.

The assignment consists of the free response questions below.

1.1 Submission

Submit a single R Markdown file (`.Rmd`) through Canvas. The file must render to PDF; knit it yourself before you upload it. A template that contains the questions with an answer block under each part, `assignment3_template.Rmd`, is available on the course website alongside this handout. Use it, and save it as `assignment3_lastname.Rmd`.

Write mathematics in LaTeX between dollar signs. Where you use R, put the code in a chunk so that your reader can see both the code and the output you rely on. If you include an image, upload it together with the `.Rmd`, since your file must render from exactly what you submit. No R package is required for this assignment; if you load any, do so in the setup chunk at the top of the template.

2 Free Response Questions

Answer each of the following. Where a question has parts, answer every part. Show your reasoning; a correct conclusion with no argument behind it will not receive useful feedback from your peer reviewer. Questions 1–2 cover Week 5; Questions 3–4 cover Week 6.

2.1 Question 1: Two Coding Standards in One Model

An analyst is trying to study whether neighborhood tree-canopy cover (continuous, in percentage points of land area) affects a continuous depressive-symptom score, using a linear model. The confounders are age (years), household income (six ordered bands, coded 1 through 6 in the data file), and census region (four regions, coded 1 through 4). The first regression model enters every variable as the numeric variable in the data file (i.e., no dummy variables, no splines, no polynomials, etc):

$$E(Y \mid \text{canopy, age, income, region}) = \beta_0 + \beta_1 \text{canopy} + \beta_2 \text{age} + \beta_3 \text{income} + \beta_4 \text{region}.$$

- Split the right-hand side into its target and nuisance functions, and state what each one is coded *for*.
- Region is entered as a single numeric variable. What does that coding assert about the outcome mean across the four regions? Is this a reasonable assertion? What does the fix cost in parameters? Income

is also a single numeric variable; say in one sentence why the same objection applies less strongly for the income variable.

- c) The canopy coefficient is 0.004 symptom-score points per percentage point of canopy, with standard error 0.002, and a reviewer calls it “negligible.” Report the coefficient and standard error per 10 percentage points and say what happens to the Wald test statistic (defined as the point estimate / standard error). A colleague proposes a standardized effect, rescaling canopy to standard-deviation units instead. The SD of canopy cover is 12 points in this cohort. In a separate replication cohort, the corresponding SD is 20. Why would the two cohorts’ per-SD coefficients differ even if the canopy effect was identical?

2.2 Question 2: Coordinates or Filter?

A closed cohort is cross-classified by a binary exposure X , a binary covariate C , and a binary outcome Y , with 100 people in each (X, C) cell, with cell numerators representing the event tally:

	$X = 0$	$X = 1$
$C = 0$	10 / 100	20 / 100
$C = 1$	30 / 100	40 / 100

- a) Compute the risk difference, risk ratio, and odds ratio for X within each level of C , and say which contrast is the same in both strata. Then consider the saturated binomial model $g[P(Y = 1 | X, C)] = \beta_0 + \beta_1 X + \beta_2 C + \beta_3 XC$ under the identity, log, and logit links. Without fitting anything, say what the fitted risks in the four cells are in terms of model parameters, write the four identity-link coefficients by hand. What does the link determine in a saturated model?
- b) Now drop the product term, leaving three parameters for four cells. Name the one pattern each link’s model now keeps (one risk difference, one risk ratio, or one odds ratio). Exactly one of the three reproduces all four observed risks: say which, and explain how a three-parameter model can be lossless here. For the other two, say what feature of the table ends up in the residuals. (No fitting is needed; in R you would find fitted risks of 0.122 / 0.179 / 0.282 / 0.415 under the log link and 0.114 / 0.186 / 0.286 / 0.414 under the logit.)

2.3 Question 3: Choosing the Family Is Choosing the Variance

An analyst models an indicator of above-median weight gain (Y) against quitting smoking (X) with a log link, so that $\exp(\beta_1)$ is a risk ratio, under three families with the same right-hand side.

- *Crude model* (X only): risks are 0.467 among non-quitters and 0.596 among quitters, and the binomial, Poisson, and Gaussian families return the same risk ratio, 1.275517, to six decimals.
 - *Adjusted model* (X plus age, sex, race, exercise, marital status, and smoking intensity): the three families return 1.333, 1.355, and 1.348. The model-based standard error of $\hat{\beta}_1$ is 0.050 under the binomial and 0.080 under the Poisson; a robust (sandwich) correction of the Poisson fit gives 0.052, with $\hat{\beta}_1$ unchanged.
- a) Why are the three crude estimates identical, and the three adjusted estimates close but not identical? Say what the family contributes to the point estimate, and why that contribution disappears in a saturated model.
- b) The Poisson model-based standard error is 60% larger than the binomial’s. Using the two variance functions, μ and $\mu(1 - \mu)$, explain the direction of the error, and say which direction it would take instead for a count outcome whose variance exceeds its mean.

2.4 Question 4: Two Ways a GLM Fails to Converge

- a) An analyst fits a log-binomial model, `family = binomial(link = "log")`, to a binary outcome and receives the message *"no valid set of coefficients has been found: please supply starting values."* Why can the binomial likelihood not be evaluated there? Why can a logistic model never produce this message?
- b) With starting values supplied, the fit converges (risk ratio 1.33, largest fitted probability 0.85). Give one reason to treat such a rescued fit with caution. Name two routine alternatives, and detail how you would apply them to data.
- c) In a 12-site cohort study that adjusts for site as a nuisance factor, one site enrolled 9 patients, all of whom had the outcome. The logistic fit reports a coefficient of 17.5 for that site's indicator, with standard error 789, and fitted probabilities of 1.000 for those 9 patients. Name this phenomenon, and give two remedies to the problem discussed (or to be discussed) in class. Say in one sentence how this failure differs from the one in (a).